Bassez 2021 Breast Cancer — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10, n_pcs=30) on sample, swept {2,5,10,20} × n_pcs {20,30,40} on a per-neighbourhood iLISI: median 4.9→8.6 of 41 samples (worst-neighbourhood p10 1.2→3.2), plateaus 2→10 then declines at θ=20. One connected component; 418 speck cells (0.63%) in 38 islands removed. Samples enter as fine intermixed speckle across one continuous, low-cycling body (3.1% cycling) — no per-patient archipelago → GO.

Gate 2 — module geography → GO (non-branching)

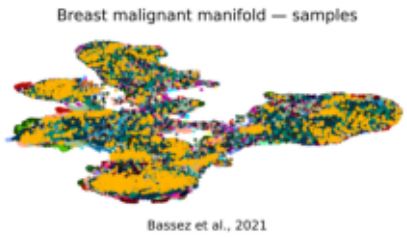
Single (Gast/pSom-origin) → Crest-terminus gradient. Smoothed geography: Gast~pSom −0.20, Crest~pSom −0.82, Crest~Gast −0.24. Along pseudotime pSom falls (ρ−0.27) and Crest rises (ρ+0.27); joint pSom&Crest enrichment 0.23 (<1 = inverse terminal gradient, not a co-expression hub). No Gast-enriched arm at any PPT stiffness. Crest-arm inversion NOT flagged (Gast&Crest enrichment 0.50 <1; Crest tracks the terminus).

Gate 3 — hub cross-repression → NA

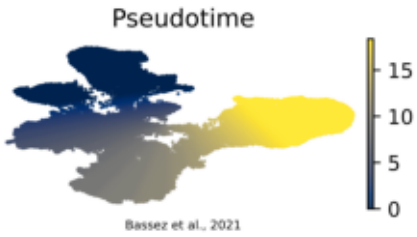
Non-branching topology (curve: 2 tips / 0 forks across N=15/20/30; PPT tree forks lack a Gast arm — max-tip Gast −0.001 vs Crest~pSom −0.87..−0.91, i.e. lobe over-fitting of one axis). No bifurcation to test → NA.

STEP FIGURES

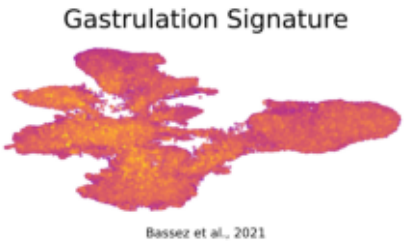
Manifold — samples



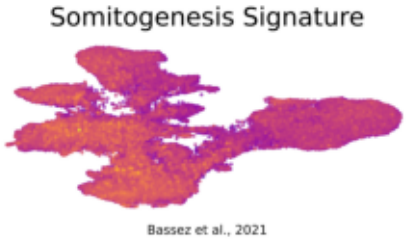
Pseudotime



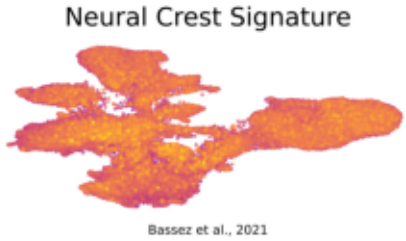
Gastrulation



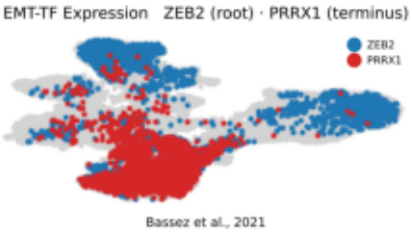
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Bassez et al., 2021), 10x — 65,885 malignant cells (42 samples); 65,464 analysed (41 samples, cell floor 17).

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching); no Gast→pSom bifurcation, so Gate 3 is NA. Geometry Linear-pSom/Crest, consistent with the linear outcome across most CCCA datasets. Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw means Gast −0.014 / pSom +0.022 / Crest −0.005. EMT-TFs ZEB2 (early/root) → PRRX1 (late/terminus).

Parameter choices

Threshold	17 cells/sample (geom mean across 92→3 spillover cliff); dropped BIOKEY_34 → 41/42 samples	**Embedding**	PCA 50, no sc.pp.scale; Harmony θ=10; n_pcs=30, n_neighbors=30, UMAP spread=1.3; 2,498 overdispersed
Trajectory	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.43); continuity corr=+0.93	**Root**	tip 2 = Gast/pSom-high, Crest-low (origin; Crest LATE); pseudotime vs cycling +0.17
Geometry	Linear-pSom/Crest; raw means Gast −0.014 / pSom +0.022 / Crest −0.005	**EMT-TFs**	ZEB2 (early/root) → PRRX1 (late/terminus), detection-gated